

AI in Action: Algorithmic Learning with Strategic Consumers

Stephan Waizmann ^{*†}

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This paper investigates the impact of artificial intelligence on the interaction between firms and consumers. It focuses on the use of learning algorithms in environments with strategic consumers — where learning must occur in the face of consumers who best-respond and adapt their behavior. An algorithm is transparent if consumers observe its inputs, whereas it is opaque if consumers do not observe its inputs. Opaque algorithms perform better than transparent ones. In contrast to a transparent algorithm, an opaque algorithm learns the optimal policy, and maximizes long-run profits. Opaque algorithms outperform transparent ones in terms of consumer welfare in important applications. That is, consumers may benefit from having less information about the algorithm’s inputs. Whether the firm benefits from using an algorithm instead of behaving strategically depends on consumers’ information about the algorithm’s inputs. When the algorithm is opaque, it yields higher payoffs than a fully strategic firm.

^{*}Department of Economics, Yale University, stephan.waizmann@yale.edu.

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